

Example of SCL file for IEC 61850 communication of the CCI, according to CEI 0-16 Annex T

Esempio di file SCL per la comunicazione IEC 61850 del CCI, secondo
CEI 0-16 Allegato T



TECHNICAL REPORT

Abstract

This technical report proposes an example of a CID file of configuration of the Central Plant Controller in accordance with the requirements expressed in Annex T of CEI 0-16 standard, regarding the mandatory performance (observability) and optional (controllability) functional performances.

CEI IDENTIFICATION DATA

Italian Standard CEI TR 57-126

Categorization CEI 57-126

LINKS/RELATIONSHIPS BETWEEN DOCUMENTS

Nationals PRJ-CEI 3515

Europeans

International

Reading Key

PUBLISHING INFORMATION

Document Type TECHNICAL REPORT

Publication Status In Force

Effective Date 01-11-2022

Issue Number 25413

Previous Edition Issue

Technical Committee CT 57 - Power systems management and associated information exchange

Identical to

Approved by President of CEI

On Date 10-10-2022

Subjected to

Close on Date

ICS 33.040.60



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Summary

This TR proposes an example of a CID file of configuration of the Central Plant Controller in accordance with the requirements expressed in Annex T of CEI 0-16 standard, regarding the mandatory performance (observability) and optional (controllability) functional performances.



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1 Introduction

This TR offers guidance on the engineering of the Central Plant Controller and its application to a Use Case (an example of a plant with a well-defined internal structure and meeting the requirements of IEC 0-16), arriving at the realization of the SCL file of the CPC configuration.

The functional performances covered by this TR are those that are mandatory (Observability) and optional ones (Controllability).

2 Scope

CEI 0-16 has chosen, among various available technologies, the use of the IEC 61850 standard for the communication between CPC and DSO1. Since this standard is very broad in its structure and in its applicability to various areas of the (electrical) power world, we aim to provide a practical example of implementation of the Data Model and Communication in compliance with Annex O of CEI 0-16:2022-03 and Annex T of CEI 0-16;V3.

This TR is intended to facilitate the implementation uniformity of the IEC 61850 interface at the CPC terminals for the purpose of high interoperability between IEDs CPC with the role of IEC 61850 Server on the DER plant side, IEC 61850 Client on the DSO side of different manufacturers, present in DER plants and DSO ¹⁾ substations, minimization of potential errors/implementation gaps due to possible interpretations of the national standard.

It is also believed that this TR can be useful in a technical discussion among the parties involved in the implementation of Smart Grid systems connected to the MV grid.

¹⁾ Annex T refers to the possible use of the IEC 61850 standard also for communication between CPCs and other Enabled Remote Actors.



3 Terms and definitions

3.1

Configured IED Description (CID)

SCL file of engineering an IED with the instantiation of both the data models and the communication for the commissioning of the device

3.2

Central Plant Controller (CPC)

Apparatus whose main functions are plant monitoring, data exchange between the plant and the DSO and any additional enabled remote actors, as well as the regulation and control of the plant itself

3.3

Data Object (DO)

Part of an LN object that represents specific information, e.g., status or measurements

3.4

Distribution System Operator (DSO)

Electric power distribution utility

3.5

Adjustment Function (AF)

CPC function related to PF2

3.6

Intelligent Electronic Device (IED)

Any device that incorporates one or more processors with the ability to receive/send data/control from/to an external source (e.g., multifunction electronic meters, relays digital, controllers)

3.7

Interoperability

Ability of two or more FDI from the same or different suppliers to exchange information and use that information for the proper execution of specific functions

3.8

Logical Node (LN)

The smallest part of a function that exchanges data. An LN is an object defined by its data and methods

3.9

Logical Device (LD)

Virtual device that exists to allow aggregation of related logical nodes

3.10

Physical Device (PD)

equivalent to an FDI as used in the context of the IEC 61850 standard

3.11

Functional Performance (FP)

Set of functions aggregated according to the classification proposed in Annex O: Mandatory (FP1 - Observability), Optional (FP2 - Controllability), Optional (FP3 - MSD)

3.12

Single Generation Group (SGG)

Generator whose Nominal Power exceeds the observability threshold provided in O.8.4

3.13

Substation/System Configuration Language (SCL)

System (formerly Substation) Configuration description Language is the format specified by IEC 61850 for the configuration of electrical station/plant devices and systems.



4 Normative references

CEI Standard 0-16 “Reference technical rule for the connection of active and passive users to the HV and MV networks of power distributors” (current version).

IEC series 61850 “Communication networks and systems for power utility automation”

5 The engineering of the CPC according to Annex T to CEI 0-16

5.1 General Information

The description of the example plant to which to apply what is discussed in the sections of this Chapter can be found in Chapter 6 (with reference also to Figure 1 and Figure 2).

The CPC, as a server, implements the IEC 61850 Ed.2.0 standard and related SCL and scheme (reference to T.3.2.1).

The name of the CPC, to be used within the CID file, will be assigned according to a convention agreed between the DSO and the organization to which the plant belongs and respecting the constraints provided by the IEC 61850 scheme (length, characters allowed), shall be unique for the client of the DSO to which it connects and thus allow unambiguous information exchange at all levels of the Data Model.

5.2 The CPC Data Model

The CPC Data Model, for all functional benefits/performances (mandatory and optional) is implemented by instantiating the data classes provided in T.3.3.1 (and sub-clauses) and, as prescribed, will be organized within a single Logical Device LD_Plant.

Having Annex T already defined the prefixes and distinguished the types of LNs in order to highlight the different DO compositions within LNs of the same class, in the next paragraphs it is considered useful to evidence only those sections of the CID file that require specialization by the configurator²⁾ based on the structural/construction characteristics of the individual plant.

The parts that are, on the other hand, common for any installation, thus already well defined in Annex T, will be directly implemented in the CID file without further description.

5.2.1 Functional Performance PF1 (Observability)

The Data Model concerning the Observability is based on the classes prescribed in T.3.3.1.1 and relevant sub-clauses.

5.2.1.1 FDI License Plate Information

With reference to T.3.3.1.1.2, the configurator shall:

- LPHD/PhyNam/Vendor -> enter the name of the Manufacturer of the CPC,
- LPHD/PhyNam/swRev -> enter the SW Version of the CPC,
- LPHD/PhyNam/location -> enter the Identifier of the CP (POD).

5.2.1.2 Operating characteristics at the point of connection (vector of characteristic powers)

With reference to T.3.3.1.1.3, the configurator shall:

- CP_WiDPCC/WRtg/setMag/f -> enter the value of the maximum active power in input (kW) of the plant,
- CP_WaDPCC/WRtg/setMag/f -> enter the value of the maximum active Power in absorption (kW) of the plant,

²⁾ The TR does not go into the type/technology/nature of the configurator adopted by the CPC manufacturer.



- CP_QiDPCC/VARtg/setMag/f -> enter the value of the maximum inductive reactive power (kVAr) of the plant,
- CP_QcDPCC/VARtg/setMag/f -> enter the value of the maximum capacitive reactive power (kVAr) of the plant,
- CP_VADPCC/VARtg/setMag/f -> enter the value of the maximum apparent power (kVA) of plant .

5.2.1.3 Phase Current Measurements at the CP at 4 sec.

With reference to T.3.3.1.1.7, the configurator may

- enter in the LN type MMXU1 dedicated to plant measurements at the delivery point the <DO name="A" type= "WYExxx" />

if the plant will make this optional information available to the DSO.

5.2.1.4 Measurements of P for Single Generation Group at 4 sec.

With reference to T.3.3.1.1.7, the configurator shall

- enter into the CID file a number N (1..99) instances (inst="1"... "N") of the LN SGGMMXU, one

for each SGG whose size (Active Power) is significant for observability purposes.

5.2.1.5 Plant Operating Status - Single Generation Group

With reference to T.3.3.1.1.9, the configurator shall:

- enter into the CID file a number N (1..99) of instances (inst="1"... "N") of the LN SSGGDGEN, one

for each SINGLE GENERATION GROUP whose size (Active Power) is significant for observability purposes;

- SSGGDGEN/GnGrId/stVal -> enter N, Identifier of the SINGLE GENERATION GROUP, equal to the instance number of the LN that includes it.

NOTE The value N must be the same as the one used in 5.2.1.4 so that the information of Operational Status and P measurements are consistently referred to the same SINGLE GENERATION GROUP observed.

5.2.2 PF2 Functional Performance (Controllability)

The Data Model relevant to Controllability is based on the classes prescribed in T.3.3.1.2 and Sub-clauses.

The Controllability Functions are optional, however the CPC will have the Data Model capability able of instantiating all of them. According to the Operating Rules (Annex U), the DSO will be aware of only those FRs that can be enabled and parameterized during the plant control process.

For each i-th Regulatory Function (FRi) among the regulated ones:

- 1) active power limitation (reference T.3.3.1.2.1),
- 2) modulation of active power input/absorption to the CP (reference T.3.3.1.2.2),
- 3) voltage regulation with delivery of reactive Inductive/Capacitive power (reference T.3.3.1.2.3),
- 4) power factor set point (reference T.3.3.1.2.4),
- 5) Q(V) (reference T.3.3.1.2.5),
- 6) $\cos\phi = f(P)$ (reference T.3.3.1.2.6),



the configurator will need to:

- FRI/Beh -> enter the default value of the operational status (1 (on) = Operational, 5 (off) = Non-operational);
- FRI/Mod -> enter the default value of the operating mode activation status (1 (on) = Active, 5 (off) = Inactive);
- FRI/Setpoint-specific -> enter the default value applied to FRI activation until re-parameterization by the DSO;

In illustrative terms, active power modulation will be considered available in input/absorption to the CP. Of this we will highlight the aspects regarding control (DSO) and engineering (configurator).

5.3 The communication of the CPC according to Annex T to CEI 0-16

Annex T is eminently focused and in detail on data modeling yet keeping conceptual levels with regard to

- protocol services (reference T.3.3.2),
- performance classes and data exchange modes (reference T.3.2.1)
 - Requests (reading/writing) by the client directly,
 - Spontaneous reporting on variations,
 - Periodic reporting,

suitable for information exchange between the CPC and the DSO³⁾ without going into the details (for generality) about how the data are organized inside the messages on the communication interface.

In this document, however, also this aspect is specified and implemented by taking cue from the specifications expressed in T.3.3.1.4, Table 98.

5.3.1 Communication section

The IP addressing of the CPC, to be used within the CID file, will be statically assigned according to an addressing plan assigned by the DSO; it will have to be unique to enable an unambiguous information exchange at all levels of the data model.

Alternatively, the DSO may offer a DHCP service for dynamic assignment of the address of the CPC.

The sub-network bitrate will reflect the characteristics of the dedicated communication port (reference O.13.1.1.1) and the communication infrastructure to which the device is connected.

5.3.2 Information flows from/to the DSO

This information exchange takes the form of a Client/Server interaction based on communication services whose protocol stack rests on MMS over TCP/IP (with reference to T.3.3.3).

5.3.2.1 Reporting to the DSO (Client)

With reference to T.3.3.2, the Server (CPC) implements reporting towards the Client (of the DSO) involving both Buffered and Unbuffered messages. The following conventions are adopted:

- Data Set for a Report - has the prefix "DS_R_" followed by the descriptive name of the Data Set;

³⁾ And additional possible Remote Enabled Actors as mentioned in Note 1.



- Control Block for Buffered Report - has the prefix "brcb_" followed by the descriptive name of the Data Set,
- Control Block for Unbuffered Report - has the prefix "urcb_" followed by the descriptive name of the Data Set,
- Data Set and Control Block are defined within LLN0.

Starting from T.3.3.1.4, Table 98, the following implementations take place

- to communicate Alarm Status and CPC Signals.
 - DataSet DS_R_Status_Alarms_Signals;
 - add an FCDA related to SSGGDGEN/Health (with InInst = N) for each Individual Generation Group;
 - this example also includes the operational status and the activation status, operating modes of each Adjustment Function;
 - the related Buffered Report brcb_Status_Alarms_Signals with transmission causes DataChange, Quality Change, and General Interrogation;
- to communicate Measurements to the CP at 4sec of the CPC
 - DataSet DS_R_CP_Mis4sec; optionally add an FCDA for "A" (current measurements);
 - the related Unbuffered Report urcb_CP_Mis4sec with transmission causes periodicity

and General Interrogation;

- to report 4sec Measurements per Generation Source and Accumulation of the CPC
 - DataSet DS_R_GenAcc_Mis4sec; measurements, by generation source, are present or not depending on the type of Generators installed in the plant: only the implemented sources will be included in the DataSet;
 - the related Unbuffered Report urcb_GenAcc_Mis4sec with causes of transmission periodicity and General Interrogation.
 - to communicate 4sec Measurements per Single Generator of the CPC
 - DataSet DS_R_SingGen_Mis4sec; add a couple of FCDAs [SGGMMXU/TotW + SSGGDGEN/GnGrId] (with InInst = N) for each Individual Generation Group;
 - the related Unbuffered Report urcb_SingGen_Mis4sec with causes of transmission
- Periodicity and General Interrogation.

NOTE As expressed in 5.2.1.4 and 5.2.1.5, the N value shall be that used in the Operational Status model and Measurements of P so that they are consistently referred to the same SINGLE GENERATION GROUP observed.

5.3.3 Commands and Configurations to the CPC (Server)

With reference to T.3.3.2, the Client (of the DSO) implements a communication to the Server (CPC) allowing direct access to specific parts of the model (LD, LN, DO, DA) as well as to structures such as the Control Block, DataSet, etc. as prescribed by the standard for the specific information element and according to access privileges defined by Annex T.



In the example of the CPC of this document, the DSO Client will be able to:

- to read values (e.g. the IED Information Label);
- to carry out SBOw type control operations, thanks to the IEDs remote location and, for better (functional) security, to access sensitive CPC resources;
- to run calibrations of functional parameters.

6 Use Case of Plant and relevant CPC

6.1 Description of the Use Case

In order to give evidence of the Data Models and of the Communication from/to the DSO on the base of the Plant characteristics, we assume the presence of the following information/features.

(in order, as represented in Figure 1 and Figure 2).

1) Observability

- CPC information label;
- Characteristic power vector (polygonal curve) of the plant;
- Information about the availability to implement adjustments; for PF1
 - at plant level = Not available,
 - of the generation macro-block = Not available,
 - of the storage macro-block = Not Available;
- DG for which the switch position is sent;
- two photovoltaic generators
 - GenPV (1) of proper size for Observability (reference CEI 0-16, O.8.4), for which the single Generator measurements are sent,
 - operational status,
 - the identifier = 1;
 - GenPV (2) under size proper for Observability (reference CEI 0-16, O.8.4), which contributes to the aggregate measurement for the source "PV" only;
- two thermoelectric generators
 - GenTher (1) of proper size for Observability (reference CEI 0-16, O.8.4), for which the single Generator measurements are sent,
 - operational status,
 - the identifier = 2;
 - GenTher (2) under size proper for Observability (reference CEI 0-16, O.8.4), which contributes to the aggregate measurement for the source "Ther" only;
- A Hydroelectric Generator GenHydr(1) under size proper for Observability (reference CEI 0-16, O.8.4), contributing to the aggregate measurement for "Hydr" source only;
- No Wind Generator;
- A Storage system under size proper for Observability (reference CEI 0-16, O.8.4), contributing to the aggregate measurement for "St" source only;
- "CP" measurements at the Connection Points.

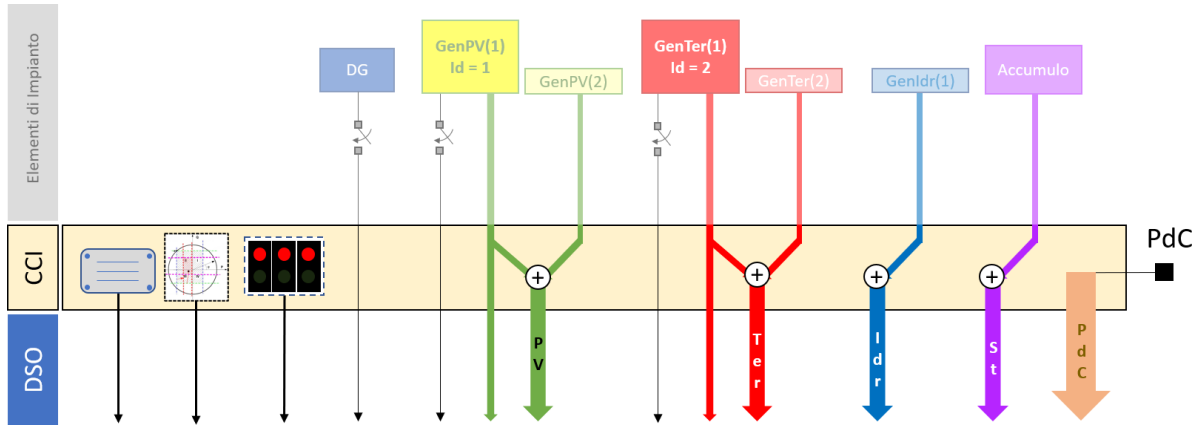


Figure 1 – Plant scheme for PF1

2) Controllability

- Information about availability to implement adjustments; for PF2
 - at plant level = Available,
 - of the generation macro-block = Available,
 - of the storage macro-block = Available;
- All the adjustment functions provided in Annex O which, each with their own characteristics, contemplate information exchanges for command / set -point / status. Only the modulation function of active power in input/absorption at the CP (Set point of active power from DSO) is available.

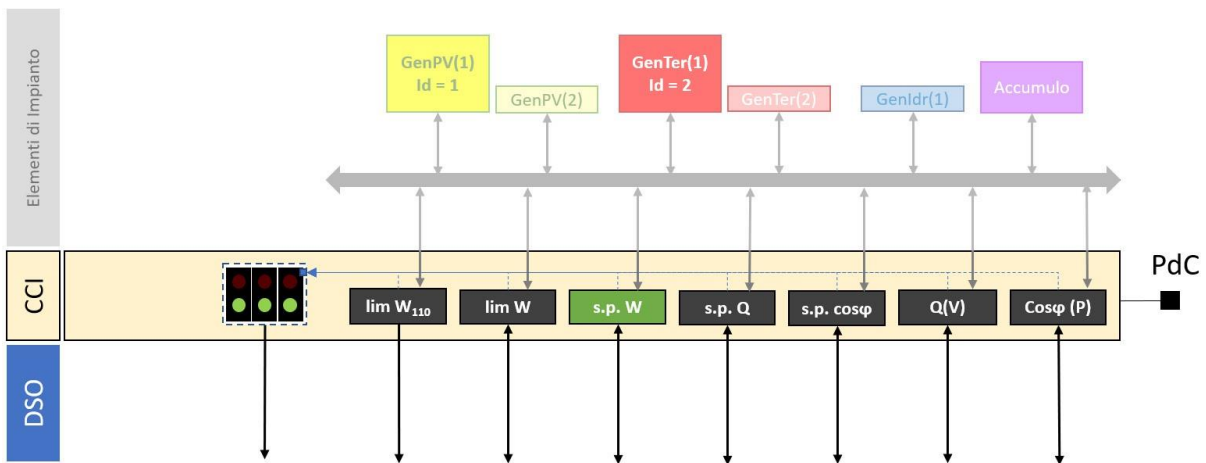


Figure 2 – Plant scheme for PF2



6.2 Example of Configurations

The following Table 1 shows the example values for configuring the CPC related to the Use Case.

Table 1 - Example of parameterizations

Parameter/Information	Value	NOTES
FDI name	CCI016_01	
IP address	192.168.8.167	
netmask	255.255.255.0	
default gateway	192.168.8.1	
Subnetwork description	CCI-DSO bus	
Vendor	Abcd_01	
swRev	1.0	
Location	IT000E123456789	POD
Maximum active power input	200	kW
Maximum active power absorption	200	kW
Maximum inductive reactive power	50	kVAr
Maximum capacitive reactive Power	50	kVAr
Maximum apparent plant power	210	kVA
Availability for regulation at plant level	1 (on) = Available	
Availability for regulation at generation macro-block level	1 (on) = Available	
Availability for regulation at storage macro-block level	1 (on) = Available	
The system provides 4-sec measurement of the Currents at the PoC	Not applicable	Optional measure according to T.3.3.1.1.7 and introduced in the Data Model of this example
The plant provides the P-Measurement of the Single Photovoltaic Generation Group GenPV(1)	GnGrId =1	
The plant provides the P-Measurement of the Single Thermoelectric Generation Group GenTer(1)	GnGrId =2	



The plant provides the Operational Status of the Single Photovoltaic Generation Group GenPV(1)	GnGrId =1	
The plant provides the Operational Status of the Single Thermoelectric Generation Group GenTer(1)	GnGrId =2	
Operating status active power limitation (Wlim)	5 (off) = Not operative	The expected values of FctOpStAuto and FctOpSt will be 'Not available'.
Wlim activation	5 (off) = Inactive	
Operating status of active power input/output modulation at the PdC (WSd)	1 (on) = Operative	The expected value of FctOpSt will be "Served
WSd activation	1 (on) = Active	
Default nominal value WSptPct	20	(percentage, with sign, in relation to the maximum apparent power of the system Smax)
Operating state voltage regulation with inductive/capacitive reactive power supply (VArSd)	5 (off) = Not operative	The expected value of FctOpSt will be 'Not available'.



Parametro/Informazione	Valore	NOTE
VARsd activation	5 (off) = Inactive	
Operational Status (PFSP)	5 (off) = Not operative	The expected value of FctOpSt will be 'Not available'.
PFSP Activation	5 (off) = Inactive	
Operational Status Q(V) (VARV)	5 (off) = Not operative	The configuration of Beh is propagated to the LN instances DPMC and DECP which model, as a whole, the FR. The expected value of FctOpSt will be 'Not Available'.
VARV activation	5 (off) = Inactive	
Operating state $\cos\phi = f(P)$ (PFW)	5 (off) = Not operative	The expected value of FctOpSt will be 'Not Available'.
PFW activation	5 (off) = Inactive	

Annex A CC SCL file

Example of CCI CID file implementing, according to IEC 61850 SCL, the Use Case described in Chapter 6.

```
<?xml version="1.0" encoding="UTF-8">
<SCL xmlns:scl="http://www3.org/2001/XMLSchema-instance" xmlns:xy="http://www.iec.ch/61850/2003/SCL/coordinates" xsi:schemaLocation="http://www.iec.ch/61850/2003/SCL SCL.xsd" version="2007" revision="B" release="4" xmlns="http://www.iec.ch/61850/2003/SCL">
  <Header id="CCI_Oss_Contr" version="1" revision="1" toolID="" nameStructure="IEDName">
    <History>
      <Item version="1" revision="1" when="05/08/2022 12:00:00" what="Esempio CID CCI Osservab. + Controllab." who="RMS" why="Per TR 57-126 relativo a Allegato T CEI 0-16" />
    </History>
  </Header>
  <Communication>
    <SubNetwork name="subnetworld" type="B4MMS" desc="">
      <Text>CCI-DSO bus</Text>
      <BitRate unit="bits">100000</BitRate>
      <ConnectedAP iedName="CCI016_01" apName="accessPoint1">
        <Address>
          <P type="IP">192.168.8.167</P>
          <P type="IP-SUBNET">255.255.255.0</P>
          <P type="IP-GATEWAY">192.168.8.1</P>
          <P type="OSI-TSEL">0001</P>
          <P type="OSI-HSEL">00000001</P>
          <P type="OSI-SSEL">0001</P>
          <P type="OSI-AP-Title">000</P>
          <P type="OSI-AE-Qualifier">000</P>
        </Address>
      </ConnectedAP>
    </SubNetwork>
  </Communication>
  <IED name="CCI016_01" engRight="full" originalSclRevision="B" originalSclVersion="2007">
    <Services nameLength="32">
      <DynAssociation />
      <GetDirectory />
      <GetDataObjectDefinition />
      <GetDataSetValue />
      <DataSetDirectory />
      <ConfDataSet max="10" maxAttributes="200" modify="false" />
      <ReadWrite />
      <ConfReportControl max="10" bufMode="both" bufConf="false" />
      <GetCBValues />
      <ConfLogControl max="1" />
      <GOOSE max="5" />
      <FileHandling />
      <GSEDir />
      <TimerActivatedControl />
      <ClientServices bufReport="true" goose="true" readLog="true" unbufReport="true" supportsLdName="true" maxAttributes="200" maxReports="10" maxGOOSE="5" />
      <DataObjectDirectory />
      <ReportSettings cbName="Fix" daSet="Fix" pitID="Fix" opFields="Fix" bufTime="Fix" tgOps="Fix" intgPd="Fix" resTms="false" owner="false" />
      <GSESettings cbName="Fix" daSet="Fix" applID="Fix" />
      <ConfLNs fixPref="true" fixLnInst="true" />
      <ConfLdName />
      <SupSubscription maxGo="10" maxSv="0" />
      <TimeSyncProt snp="true" c37_238="false" other="true" />
    </Services>
    <AccessPoint name="accessPoint1" desc="">
      <Server desc="CCI CEI 0-16 Oss. e Contr.">
        <Authentication />
        <LDDevice inst="LD_Plant" desc="">
          <LNO lClass="LLN0" lType="LLN01" inst="">

```



```

<DataSet name="DS_R_Stato_Allarmi_Segnali" desc="Stato Allarmi e Segnali del CCI">
  <FCDA Icdns="LD_Plant" InClass="DECP" Icd="ST" InInst="1" Prefix="DisFR" doName="Beh1" />
  <FCDA Icdns="LD_Plant" InClass="DGEN" Icd="ST" InInst="1" Prefix="DisFR" doName="Beh1" />
  <FCDA Icdns="LD_Plant" InClass="DSTO" Icd="ST" InInst="1" Prefix="DisFR" doName="Beh1" />
  <FCDA Icdns="LD_Plant" InClass="XGBR" Icd="ST" InInst="1" Prefix="IDG" doName="Pos" />
  <FCDA Icdns="LD_Plant" InClass="DGEN" Icd="ST" InInst="1" Prefix="SSGG" doName="Health" />
  <FCDA Icdns="LD_Plant" InClass="DGEN" Icd="ST" InInst="2" Prefix="SSGG" doName="Health" />
  <FCDA Icdns="LD_Plant" InClass="DWMX" Icd="ST" InInst="1" Prefix="Wlim" doName="Beh1" />
  <FCDA Icdns="LD_Plant" InClass="DWMX" Icd="ST" InInst="1" Prefix="Wlim" doName="FctOpStAuto" />
  <FCDA Icdns="LD_Plant" InClass="DWMX" Icd="ST" InInst="1" Prefix="Wlim" doName="FctOpStEx" />
  <FCDA Icdns="LD_Plant" InClass="DAGC" Icd="ST" InInst="1" Prefix="WSd" doName="Beh1" />
  <FCDA Icdns="LD_Plant" InClass="DAGC" Icd="ST" InInst="1" Prefix="WSd" doName="FctOpSt" />
  <FCDA Icdns="LD_Plant" InClass="DVAR" Icd="ST" InInst="1" Prefix="VArSd" doName="Beh1" />
  <FCDA Icdns="LD_Plant" InClass="DVAR" Icd="ST" InInst="1" Prefix="VArSd" doName="FctOpSt" />
  <FCDA Icdns="LD_Plant" InClass="DPFP" Icd="ST" InInst="1" Prefix="PFSP" doName="Beh1" />
  <FCDA Icdns="LD_Plant" InClass="DPFP" Icd="ST" InInst="1" Prefix="PFSP" doName="FctOpSt" />
  <FCDA Icdns="LD_Plant" InClass="DVAR" Icd="ST" InInst="1" Prefix="VArV" doName="Beh1" />
  <FCDA Icdns="LD_Plant" InClass="DVAR" Icd="ST" InInst="1" Prefix="VArV" doName="FctOpSt" />
  <FCDA Icdns="LD_Plant" InClass="DPFW" Icd="ST" InInst="1" Prefix="PPW" doName="Beh1" />
  <FCDA Icdns="LD_Plant" InClass="DPFW" Icd="ST" InInst="1" Prefix="PPW" doName="FctOpSt" />
</DataSet>
<DataSet name="DS_R_PdC_Mis4sec" desc="Misure al PdC a 4sec del CCI">
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="1" Prefix="PdC" doName="TotW" />
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="1" Prefix="PdC" doName="TotVA" />
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="1" Prefix="PdC" doName="PPV" />
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="1" Prefix="PdC" doName="A" />
</DataSet>
<DataSet name="DS_R_GenAcc_Mis4sec" desc="Misure a 4sec per Tipo Generazione e Accumulo del CCI">
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="1" Prefix="GenPV" doName="TotW" />
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="1" Prefix="GenTer" doName="TotW" />
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="1" Prefix="GenIdr" doName="TotW" />
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="1" Prefix="Sc" doName="TotW" />
</DataSet>
<DataSet name="DS_R_SingGen_Mis4sec" desc="Misure a 4sec per Singolo Generatore del CCI">
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="1" Prefix="SGG" doName="TotW" />
  <FCDA Icdns="LD_Plant" InClass="DGEN" Icd="ST" InInst="1" Prefix="SSGG" doName="GnGrid" />
  <FCDA Icdns="LD_Plant" InClass="MMXU" Icd="MX" InInst="2" Prefix="SGG" doName="TotW" />
  <FCDA Icdns="LD_Plant" InClass="DGEN" Icd="ST" InInst="2" Prefix="SSGG" doName="GnGrid" />
</DataSet>
<ReportControl name="brcb_Stato_Allarmi_Segnali" confRev="1" dataSet="DS_R_Stato_Allarmi_Segnali" rptID="CCI016_01LD_PlantLLN0.brcb_Stato_Allarmi_Segnali" buffered="true" intgPd="0" bufTime="250" desc="Stato Allarmi e Segnali del CCI">
  <TrgOps ddg="true" qchg="true" g="true" />
  <OptFields seqNum="true" timeStamp="true" dataSet="true" reasonCodes="true" dataRef="true" entryID="true" configRef="true" buOvl="true" />
  <RptEnabled max="2" desc="Client DSO" />
</ReportControl>
<ReportControl name="urcb_PdC_Mis4sec" confRev="1" dataSet="DS_R_PdC_Mis4sec" rptID="CCI016_01LD_PlantLLN0.urcb_PdC_Mis4sec" intgPd="4000" bufTime="0" desc="Misure al PdC a 4sec del CCI">
  <TrgOps period="true" g="true" />
  <OptFields seqNum="true" timeStamp="true" dataSet="true" reasonCodes="true" dataRef="true" entryID="true" configRef="true" />
  <RptEnabled max="2" desc="Client DSO" />
</ReportControl>
<ReportControl name="urcb_GenAcc_Mis4sec" confRev="1" dataSet="DS_R_GenAcc_Mis4sec" rptID="CCI016_01LD_PlantLLN0.urcb_GenAcc_Mis4sec" intgPd="4000" bufTime="0" desc="Misure a 4sec per Tipo Generazione e Accumulo del CCI">
  <TrgOps period="true" g="true" />
  <OptFields seqNum="true" timeStamp="true" dataSet="true" reasonCodes="true" dataRef="true" entryID="true" configRef="true" />
  <RptEnabled max="2" desc="Client DSO" />
</ReportControl>
<ReportControl name="urcb_SingGen_Mis4sec" confRev="1" dataSet="DS_R_SingGen_Mis4sec" rptID="CCI016_01LD_PlantLLN0.urcb_SingGen_Mis4sec" intgPd="4000" bufTime="0" desc="Misure a 4sec per Singolo Generatore del CCI">
  <TrgOps period="true" g="true" />
  <OptFields seqNum="true" timeStamp="true" dataSet="true" reasonCodes="true" dataRef="true" entryID="true" configRef="true" />
  <RptEnabled max="2" desc="Client DSO" />
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  </DAI>
  <DAI name="ctlModel">
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  </DAI>
</DOI>

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  <DAI name="stVal">
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  </DAI>
</DOI>
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  <DAI name="stVal">
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  </DAI>
</DOI>
<DOI name="NamPlt" desc="">
  <DAI name="IdNs" desc="">
    <Val>IEC 61850-7-42010</Val>
  </DAI>
</DOI>
</LN0>
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  <DOI name="PhyNam">
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  </DOI>
  <DOI name="WRtg" desc="Pdc - Potenza attiva massima in immissione (kW)">
    <SDI name="setMag" desc="">
      <DAI name="I" desc="Valore caratteristico di impianto">
        <Val>200</Val>
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<LN hClass="DPCC" hType="DPCC1" inst="1" prefix="Pdc_Wat">
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    </DAI>
  </DOI>
  <DOI name="WRtg" desc="Pdc - Potenza attiva massima in assorbimento (kW)">
    <SDI name="setMag" desc="">
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        <Val>200</Val>
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</LN>
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  <DOI name="Beh" desc="">
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    </DAI>
  </DOI>
  <DOI name="VArRtg" desc="PdC - Potenza reattiva induttiva massima (kVar)">
    <SDI name="setMag" desc="">
      <DAI name="I" desc="Valore caratteristico di impianto">
        <Val>50</Val>
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</LN>
<LN hClass="DPCC" hType="DPCCZ" inst="1" prefix="PdC_Qc">
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  <DOI name="Beh" desc="">
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  </DOI>
  <DOI name="VArRtg" desc="PdC - Potenza reattiva capacitativa massima (kVar)">
    <SDI name="setMag" desc="">
      <DAI name="I" desc="Valore caratteristico di impianto">
        <Val>50</Val>
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<LN hClass="DPCC" hType="DPCC3" inst="1" prefix="PdC_VA">
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  <DOI name="VArRtg" desc="PdC - Potenza apparente massima di impianto (kVA)">
    <SDI name="setMag" desc="">
      <DAI name="I" desc="Inserire il valore caratteristico di impianto">
        <Val>210</Val>
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<LN hClass="DECP" hType="DECP1" inst="1" prefix="DisFR">
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</DOI>
</LN>
<LN hClass="MMXU" hType="MMXU2" inst="1" prefix="GenWF">
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</LN>
<LN hClass="MMXU" hType="MMXU2" inst="1" prefix="GenTer">
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<LN hClass="MMXU" hType="MMXU2" inst="1" prefix="GenIdr">
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  </LN>
  <LN InClass="MMXU" InType="MMXU2" inst="2" prefix="SGG" desc="Misura di P Singolo Gruppo di Generazione GenTer(1)">
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  <LN InClass="XCBR" InType="XCBR1" inst="1" prefix="IDG">
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    </DAI>
  </DOI>
  <DOI name="BltOpn" desc="">
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  </DOI>
  <DOI name="BltCls" desc="">
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</LN>
<LN InClass="DGEN" InType="DGEN2" inst="1" prefix="SSGG" desc="Singolo Gruppo di Generazione GenPV(1)">
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  <DOI name="Beh" desc="">
    <DAI name="stVal">
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  </DOI>
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  <DOI name="GnGtd" desc="Numero Identificativo">
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</DOI>
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    </DAI>
</DOI>
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    </DAI>
</DOI>
<DOI name="GrGrid" desc="Numero Identificativo">
    <DAI name="stVal">
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    </DAI>
</DOI>
</LN>
<LN lnClass="DWMX" lnType="DWMX1" inst="1" prefix="Wlim" desc="Fz. di Regolazione - limitazione della W - da DSO">
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        </DAI>
    </DOI>
    <DOI name="Beh" desc="">
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        </DAI>
    </DOI>
    <DOI name="WMaxSptPct" desc="Setpoint limite di W in generazione (percentuale con segno rispetto a Smax di Impianto) [0..100]">
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            <Val>sbw-with-enhanced-security</Val>
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    <DOI name="Mod" desc="Attivazione/disattivazione funzionamento [5 = Inattiva, 1 = Attiva]">
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        </DAI>
        <DAI name="ctlModel">
            <Val>sbw-with-enhanced-security</Val>
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        <DAI name="lnNs">
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    </DOI>
    <DOI name="Beh" desc="">
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        </DAI>
    </DOI>
    <DOI name="WSptPct" desc="Setpoint modulazione W in immissione/assorbimento al PdC (percentuale con segno rispetto alla Smax di Impianto) [0..100]">
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        <SDI name="mxVal" desc="Setpoint W in immissione/assorbimento al PdC (percentuale con segno rispetto a Smax di Impianto) [0..100]">
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                <Val>20</Val>
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        </SDI>
    </DOI>
    <DOI name="Mod" desc="Attivazione/disattivazione funzionamento - [5 = Inattiva, 1 = Attiva]">
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    <Val>on</Val>
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  <DAI name="ctlModel">
    <Val>sbwWithEnhancedSecurity</Val>
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    <DAI name="InNs">
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    <DOI>
      <DOI name="Beh" desc="">
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      <DOI name="VAiTgtSptPct" desc="Setpoint VAR induttiva/capacitiva (percentuale con segno rispetto a Smax di Impianto) [0..100]">
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      </DOI>
      <DOI name="Mod" desc="Attivazione/disattivazione funzionamento [5 = Inattiva, 1 = Attiva]">
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        <DAI name="ctlModel">
          <Val>sbwWithEnhancedSecurity</Val>
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    <LN hClass="DFFF" InType="DFFF1" inst="1" prefix="PFSP" desc="Fz. di Regolazione - set point del fattore di potenza - da DSO">
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        <DAI name="InNs">
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        <DOI>
          <DOI name="Beh" desc="">
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            </DAI>
          </DOI>
          <DOI name="PFGnTgtSpt" desc="Setpoint FdP in caso di generazione di potenza attiva [-1,00..0,00]">
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          <DOI name="PFLodTgtSpt" desc="Setpoint di FdP in caso di assorbimento di potenza attiva [0,00..1,00]">
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          </DOI>
          <DOI name="Mod" desc="Attivazione/disattivazione funzionamento [5 = Inattiva, 1 = Attiva]">
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            <DAI name="ctlModel">
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            </DAI>
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        <LN hClass="DVVR" InType="DVVR1" inst="1" prefix="VAiV" desc="Fz. di Regolazione - Q(V) - da DSO">
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</DOI>
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  <DOI name="K" desc="Parametro K della funzione Q(V) [-1.00..1.00]" />
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  </DOI>
<DOI>
  <DOI name="WSpt1" desc="Potenza di Lock-in della funzione Q(V) - [0.00..max] della PNominale (P.U.)">
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<DOI>
  <DOI name="WSpt1" desc="Potenza di Lock-out della funzione Q(V) - [0.00..max] della PNominale (P.U.)">
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</LN>
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  <DOI name="Beh" desc="">
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<DOI>
  <DOI name="VMax" desc="Tensione superiore 1 della funzione Q(V) [0.00..max] della VNominale (P.U.)" />
  <DOI name="VMin" desc="Tensione inferiore 1 della funzione Q(V) [0.00..max] della VNominale (P.U.)" />
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    </DOI>
    <DOI name="Mod" desc="Attivazione/disattivazione funzionamento [5 = Inattiva, 1 = Attiva]>
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      </DAI>
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        <Val>sb-with-enhanced-security</Val>
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    </DOI>
    <DOI name="WSetA" desc="Valore P (punto A) [0.00..max] della PNominale (P.U.)" />
    <DOI name="PFSetA" desc="Valore cosfi (punto A) [-1.00..1.00]" />
    <DOI name="WSetB" desc="Valore P (punto B) [0.00..max] della PNominale (P.U.)" />
    <DOI name="PFSetB" desc="Valore cosfi (punto B) [-1.00..1.00]" />
    <DOI name="WSetC" desc="Valore P (punto C) [0.00..max] della PNominale (P.U.)" />
    <DOI name="PFSetC" desc="Valore cosfi (punto C) [-1.00..1.00]" />
    <DOI name="VLkIn" desc="Tensione di Lock-in della funzione cosfi = f(P) [1.00..1.10] della VNominale (P.U.)" />
    <DOI name="VLkOut" desc="Tensione di Lock-out della funzione cosfi = f(P) [0.90..1.00] della VNominale (P.U.)" />
  </LN>
</LDevice>
</Server>
</AccessPoint>
</IED>
<DataTypeTemplates>
  <LNodeType id="LLN01" lnClass="LLN0">
    <DO name="Mod" type="ENC_1_Mod" />
    <DO name="Beh" type="ENS_2_Beh" />
    <DO name="Health" type="ENS_1_Health" />
    <DO name="NamPit" type="LPL_5_NamPit" />
  </LNodeType>
  <LNodeType id="LPHD1" lnClass="LPHD">
    <DO name="PhyNam" type="DPL_1_PhyNam" />
    <DO name="PhyHealth" type="ENS_1_PhyHealth" />
    <DO name="Proxy" type="SPS_1_Proxy" />
  </LNodeType>
  <LNodeType id="DPCC1" lnClass="DPCC">
    <DO name="NamPit" type="LPL_5_NamPit" />
    <DO name="Beh" type="ENS_2_Beh" />
    <DO name="AreaEpsEcpld" type="VSG_2_AreaEpsEqp" />
    <DO name="AreaEpsWMax" type="ASG_2_WMaxSet" />
    <DO name="VRel" type="ASG_2_WMaxSet" />
    <DO name="WRtg" type="ASG_2_WMaxSet" />
    <DO name="RelFrm" type="SPG_2_RelFrm" />
    <DO name="EldReltd" type="VSG_2_AreaEpsEqp" />
  </LNodeType>
  <LNodeType id="DPCC2" lnClass="DPCC">
    <DO name="NamPit" type="LPL_5_NamPit" />

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<DO name="Beh1" type="ENS_2_Beh1" />
<DO name="AreaEpsEcpId1" type="VSG_2_AreaEpsEqp" />
<DO name="AreaEpsWMax1" type="ASG_2_WMaxSet" />
<DO name="VRef1" type="ASG_2_WMaxSet" />
<DO name="VArRtg1" type="ASG_2_WMaxSet" />
<DO name="RelFrm1" type="SPG_2_RelFrm" />
<DO name="ElcRelId1" type="VSG_2_AreaEpsEqp" />
</NodeType>
<NodeType id="DPCC3" hClass="DPCC">
  <DO name="NamPlt1" type="LPL_5_NamPlt" />
  <DO name="Beh1" type="ENS_2_Beh1" />
  <DO name="AreaEpsEcpId1" type="VSG_2_AreaEpsEqp" />
  <DO name="AreaEpsWMax1" type="ASG_2_WMaxSet" />
  <DO name="VRef1" type="ASG_2_WMaxSet" />
  <DO name="VArRtg1" type="ASG_2_WMaxSet" />
  <DO name="RelFrm1" type="SPG_2_RelFrm" />
  <DO name="ElcRelId1" type="VSG_2_AreaEpsEqp" />
</NodeType>
<NodeType id="DECP1" hClass="DECP">
  <DO name="NamPlt1" type="LPL_5_NamPlt" />
  <DO name="Beh1" type="ENS_2_Beh1" />
  <DO name="VRef1" type="ASG_2_WMaxSet" />
  <DO name="RelFrm1" type="SPG_2_RelFrm" />
  <DO name="ElcRelId1" type="VSG_2_AreaEpsEqp" />
</NodeType>
<NodeType id="DGEN1" hClass="DGEN">
  <DO name="NamPlt1" type="LPL_5_NamPlt" />
  <DO name="DEROpSt1" type="ENS_2_DEROpSt" />
  <DO name="Beh1" type="ENS_2_Beh1" />
  <DO name="WMaxRtg1" type="ASG_2_WMaxSet" />
  <DO name="WMaxRtg1" type="ASG_2_WMaxSet" />
  <DO name="WMax1" type="ASG_2_WMaxSet" />
  <DO name="WRmp1" type="ASG_2_WMaxSet" />
  <DO name="VMax1" type="ASG_2_WMaxSet" />
  <DO name="OutEqpRef1" type="ORG_2_EldVisRef" />
  <DO name="PhsConnTyp1" type="ENG_2_PhsConnTyp" />
  <DO name="DERTyp1" type="ENG_2_DERTyp" />
</NodeType>
<NodeType id="DSTO1" hClass="DSTO">
  <DO name="NamPlt1" type="LPL_5_NamPlt" />
  <DO name="DEROpSt1" type="ENS_2_DEROpSt" />
  <DO name="Beh1" type="ENS_2_Beh1" />
  <DO name="ElMW1" type="MV_2_CibWPct" />
  <DO name="ElMW1Pct1" type="MV_2_CibWPct" />
  <DO name="EqSto1" type="ORG_2_EldVisRef" />
  <DO name="ChalWMax1" type="ASG_2_WMaxSet" />
  <DO name="OutEqpRef1" type="ORG_2_EldVisRef" />
  <DO name="PhsConnTyp1" type="ENG_2_PhsConnTyp" />
  <DO name="DERTyp1" type="ENG_2_DERTyp" />
</NodeType>
<NodeType id="MMXU1" hClass="MMXU">
  <DO name="Beh1" type="ENS_2_Beh1" />
  <DO name="TotW1" type="MV_2_CibWPct" />
  <DO name="TotVA1" type="MV_2_CibWPct" />
  <DO name="PPV1" type="DEL_1_PPV" />
  <DO name="A1" type="WYE_1_A" />
</NodeType>
<NodeType id="MMXU2" hClass="MMXU">
  <DO name="Beh1" type="ENS_2_Beh1" />
  <DO name="TotW1" type="MV_2_CibWPct" />
</NodeType>
<NodeType id="XCBR1" hClass="XCBR">
  <DO name="Beh1" type="ENS_2_Beh1" />
  <DO name="Loc1" type="SPS_1_Proxy" />
  <DO name="OpCnt1" type="INS_1_OpCnt" />
  <DO name="Pos1" type="DPC_1_Pos" />
  <DO name="BlkOpn1" type="SPC_1_BlkOpn" />

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<DO name="BlkCls" type="SPC_1_BlkOpn" />
</NodeType>
<NodeType id="DGEN2" hClass="DGEN">
  <DO name="NamPlt" type="LPL_5_NamPlt" />
  <DO name="DEROpSt" type="ENS_2_DEROpSt" />
  <DO name="Beh" type="ENS_2_Beh" />
  <DO name="Health" type="ENS_1_Health" />
  <DO name="WMaxRtg" type="ASG_2_WMaxSet" />
  <DO name="VMaxRtg" type="ASG_2_VMaxSet" />
  <DO name="WMax" type="ASG_2_WMaxSet" />
  <DO name="WRmp" type="ASG_2_WMaxSet" />
  <DO name="VMax" type="ASG_2_WMaxSet" />
  <DO name="OutEqpRef" type="ORG_2_ElctVisRef" />
  <DO name="PhsConnTyp" type="ENG_2_PhConnTyp" />
  <DO name="DErTyp" type="ENG_2_DErTyp" />
  <DO name="GrnGrid" type="INS_1_OpCnt" />
</NodeType>
<NodeType id="DWMX1" hClass="DWMX">
  <DO name="NamPlt" type="LPL_5_NamPlt" />
  <DO name="Beh" type="ENS_2_Beh" />
  <DO name="WMaxSplPct" type="APC_1_WMaxSpl" />
  <DO name="Mod" type="ENC_2_Mod" />
  <DO name="RmpRteUse" type="SPG_2_RelFrm" />
  <DO name="InEqpRef" type="ORG_2_ElctVisRef" />
  <DO name="FctOpStAuto" type="ENS_2_FctOpSt" />
  <DO name="FctOpStIEx" type="ENS_2_FctOpSt" />
</NodeType>
<NodeType id="DAGC1" hClass="DAGC">
  <DO name="NamPlt" type="LPL_5_NamPlt" />
  <DO name="Beh" type="ENS_2_Beh" />
  <DO name="ReqW" type="MV_2_CibWPct" />
  <DO name="WSplPct" type="APC_1_WMaxSpl" />
  <DO name="Mod" type="ENC_2_Mod" />
  <DO name="RmpRteUse" type="SPG_2_RelFrm" />
  <DO name="InEqpRef" type="ORG_2_ElctVisRef" />
  <DO name="FctOpSt" type="ENS_2_FctOpSt" />
</NodeType>
<NodeType id="DVAR1" hClass="DVAR">
  <DO name="NamPlt" type="LPL_5_NamPlt" />
  <DO name="Beh" type="ENS_2_Beh" />
  <DO name="ReqVar" type="MV_2_CibWPct" />
  <DO name="VarTgtSplPct" type="APC_1_WMaxSpl" />
  <DO name="Mod" type="ENC_2_Mod" />
  <DO name="RmpRteUse" type="SPG_2_RelFrm" />
  <DO name="InEqpRef" type="ORG_2_ElctVisRef" />
  <DO name="FctOpSt" type="ENS_2_FctOpSt" />
</NodeType>
<NodeType id="DFFP1" hClass="DFFP">
  <DO name="NamPlt" type="LPL_5_NamPlt" />
  <DO name="ReqPFExt" type="SPS_1_Proxy" />
  <DO name="Beh" type="ENS_2_Beh" />
  <DO name="ReqPF" type="MV_2_CibWPct" />
  <DO name="PFGntTgtSpl" type="APC_1_WMaxSpl" />
  <DO name="PFLodTgtSpl" type="APC_1_WMaxSpl" />
  <DO name="Mod" type="ENC_2_Mod" />
  <DO name="RmpRteUse" type="SPG_2_RelFrm" />
  <DO name="InEqpRef" type="ORG_2_ElctVisRef" />
  <DO name="FctOpSt" type="ENS_2_FctOpSt" />
</NodeType>
<NodeType id="DWVR1" hClass="DWVR">
  <DO name="NamPlt" type="LPL_5_NamPlt" />
  <DO name="Beh" type="ENS_2_Beh" />
  <DO name="VRefEsp" type="MV_2_CibWPct" />
  <DO name="ReqVar" type="MV_2_CibWPct" />
  <DO name="VarTgtSplPct" type="APC_1_WMaxSpl" />
  <DO name="Mod" type="ENC_2_Mod" />
  <DO name="VArCvDel" type="CSG_1_VArCv" />

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<DO name="RmpRteUse" type="SPG_2_RelFrm" />
<DO name="InEqRef" type="ORG_2_ElctMsRef" />
<DO name="FctOpSt" type="ENS_2_FctOpSt" />
<DO name="Ik" type="ASG_2_WMaxSet" />
</NodeType>
<NodeType id="DPMC1" hClass="DPMC">
  <DO name="NamPlt" type="LPL_5_NamPlt" />
  <DO name="Beh" type="ENS_2_Beh" />
  <DO name="WSptd" type="APC_1_WMaxSpt" />
  <DO name="DERRef" type="ORG_2_ElctMsRef" />
  <DO name="OutEqRef" type="ORG_2_ElctMsRef" />
</NodeType>
<NodeType id="DECP2" hClass="DECP">
  <DO name="NamPlt" type="LPL_5_NamPlt" />
  <DO name="Beh" type="ENS_2_Beh" />
  <DO name="VRef" type="ASG_2_WMaxSet" />
  <DO name="VMax" type="ASG_2_WMaxSet" />
  <DO name="VMin" type="ASG_2_WMaxSet" />
  <DO name="RelFrm" type="SPG_2_RelFrm" />
  <DO name="ElctReltd" type="VSG_2_AreaEpsEqp" />
</NodeType>
<NodeType id="DPPW1" hClass="DPPW">
  <DO name="NamPlt" type="LPL_5_NamPlt" />
  <DO name="Beh" type="ENS_2_Beh" />
  <DO name="FctOpSt" type="ENS_2_FctOpSt" />
  <DO name="Mod" type="ENC_2_Mod" />
  <DO name="WSetA" type="ASG_2_WMaxSet" />
  <DO name="PFSetA" type="ASG_2_WMaxSet" />
  <DO name="WSetB" type="ASG_2_WMaxSet" />
  <DO name="PFSetB" type="ASG_2_WMaxSet" />
  <DO name="WSetC" type="ASG_2_WMaxSet" />
  <DO name="PFSetC" type="ASG_2_WMaxSet" />
  <DO name="VLkIn" type="ASG_2_WMaxSet" />
  <DO name="VLkOut" type="ASG_2_WMaxSet" />
  <DO name="RmpRteUse" type="SPG_2_RelFrm" />
  <DO name="InEqRef" type="ORG_2_ElctMsRef" />
</NodeType>
<DOType id="ENC_1_Mod" cd="ENC">
  <DA name="stVal" type="Mod" bType="Enum" fc="ST" dchg="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
  <DA name="ctlModel" type="CtlModelKind" bType="Enum" fc="CP" />
</DOType>
<DOType id="ENS_2_Beh" cd="ENS">
  <DA name="stVal" type="Beh" bType="Enum" fc="ST" dchg="true" dupd="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
</DOType>
<DOType id="ENS_1_Health" cd="ENS">
  <DA name="stVal" type="Health" bType="Enum" fc="ST" dchg="true" dupd="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
</DOType>
<DOType id="DPL_1_PhyNam" cd="DPL">
  <DA name="vndor" bType="VisString255" fc="DC" />
  <DA name="swRev" bType="VisString255" fc="DC" />
  <DA name="location" bType="VisString255" fc="DC" />
</DOType>
<DOType id="ENS_1_PhyHealth" cd="ENS">
  <DA name="stVal" type="PhyHealth" bType="Enum" fc="ST" dchg="true" dupd="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
</DOType>
<DOType id="SPS_1_Proxy" cd="SPS">
  <DA name="stVal" bType="BOOLEAN" fc="ST" dchg="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />

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</DOType>
<DOType id="LPL_5_NamPlt" odc="LPL">
  <DA name="vndor" bType="VisString255" fc="DC" />
  <DA name="swRev" bType="VisString255" fc="DC" />
  <DA name="idNs" bType="VisString255" fc="EX"/>
  <DA name="lnNs" bType="VisString255" fc="EX"/>
</DOType>
<DOType id="VSG_2_AreaEpsEcp" odc="VSG">
  <DA name="setVal" bType="VisString255" fc="SP" dchg="true" />
</DOType>
<DOType id="ASG_2_WMaxSet" odc="ASG">
  <DA name="setMag" type="AnalogueValue_1" bType="Struct" fc="SP" dchg="true" />
</DOType>
<DOType id="ENG_2_PhConnTyp" odc="ENG">
  <DA name="setVal" type="PhaseKind" bType="Enum" fc="SP" dchg="true" />
</DOType>
<DOType id="ENG_2_DERTyp" odc="ENG">
  <DA name="setVal" type="DERUnitKind" bType="Enum" fc="SP" dchg="true" />
</DOType>
<DOType id="ORG_2_ElomsRef" odc="ORG">
  <DA name="setSrcRef" bType="VisString129" fc="SP" dchg="true" />
</DOType>
<DOType id="SPG_2_RelFrm" odc="SPG">
  <DA name="setVal" bType="BOOLEAN" fc="SP" dchg="true" />
</DOType>
<DOType id="MV_2_CbWPct" odc="MV">
  <DA name="mag" type="AnalogueValue_1" bType="Struct" fc="MX" dchg="true" dupd="true" />
  <DA name="q" bType="Quality" fc="MX" qchg="true" />
  <DA name="t" bType="Timestamp" fc="MX" />
</DOType>
<DOType id="ENS_2_DEROpSt" odc="ENS">
  <DA name="stVal" type="DERStateKind" bType="Enum" fc="ST" dchg="true" dupd="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
</DOType>
<DOType id="DEL_1_PPV" odc="DEL">
  <SDO name="phsAB" type="CMV_1_phsAB" />
  <SDO name="phsBC" type="CMV_1_phsAB" />
  <SDO name="phsCA" type="CMV_1_phsAB" />
</DOType>
<DOType id="CMV_1_phsAB" odc="CMV">
  <DA name="cVal" type="Vector_1" bType="Struct" fc="MX" dchg="true" dupd="true" />
  <DA name="q" bType="Quality" fc="MX" qchg="true" />
  <DA name="t" bType="Timestamp" fc="MX" />
</DOType>
<DOType id="WYE_1_A" odc="WYE">
  <SDO name="phsA" type="CMV_1_phsAB" />
  <SDO name="phsB" type="CMV_1_phsAB" />
  <SDO name="phsC" type="CMV_1_phsAB" />
</DOType>
<DOType id="INS_1_QpCnt" odc="INS">
  <DA name="stVal" bType="INT32" fc="ST" dchg="true" dupd="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
</DOType>
<DOType id="DPC_1_Pos" odc="DPC">
  <DA name="stVal" bType="Dbpos" fc="ST" dchg="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
  <DA name="ctlModel" type="CIMModelKind" bType="Enum" fc="CF" dchg="true" />
</DOType>
<DOType id="SPC_1_BlkOpn" odc="SPC">
  <DA name="stVal" bType="BOOLEAN" fc="ST" dchg="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
  <DA name="ctlModel" type="CIMModelKind" bType="Enum" fc="CF" dchg="true" />
</DOType>

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<DOType id="APC_1_WMaxSpl" odc="APC">
  <DA name="Cancel" type="APCCancel_1" bType="Struct" fc="CO" />
  <DA name="Oper" type="APCOperate_1" bType="Struct" fc="CO" />
  <DA name="SBOw" type="APCSelectWithValue_1" bType="Struct" fc="CO" />
  <DA name="ctlModel" type="CtlModelKind" bType="Enum" fc="CF" />
  <DA name="mxVal" type="AnalogueValue_1" bType="Struct" fc="MX" dchg="true" />
  <DA name="q" bType="Quality" fc="MX" qchg="true" />
  <DA name="t" bType="Timestamp" fc="MX" />
</DOType>
<DOType id="ENC_2_Mod" odc="ENC">
  <DA name="stVal" type="Mod" bType="Enum" fc="ST" dchg="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
  <DA name="ctlModel" type="CtlModelKind" bType="Enum" fc="CF" />
  <DA name="SBOw" type="ENCSelectWithValue_1" bType="Struct" fc="CO" />
  <DA name="Oper" type="ENCOperate_1" bType="Struct" fc="CO" />
  <DA name="Cancel" type="ENCCancel_1" bType="Struct" fc="CO" />
</DOType>
<DOType id="ENS_2_FctOpSt" odc="ENS">
  <DA name="stVal" type="DERFunctionState" bType="Enum" fc="ST" dchg="true" dupd="true" />
  <DA name="q" bType="Quality" fc="ST" qchg="true" />
  <DA name="t" bType="Timestamp" fc="ST" />
</DOType>
<DOType id="CSG_1_WArCv" odc="CSG">
  <DA name="numPts" bType="INT16U" fc="SP" />
  <DA name="cvPts" type="Point_2" bType="Struct" count="16" fc="SP" />
  <DA name="xUnits" type="Unit_3" bType="Struct" fc="CF" />
  <DA name="yUnits" type="Unit_3" bType="Struct" fc="CF" />
  <DA name="maxPts" bType="INT16U" fc="CF" />
  <DA name="xD" bType="VisString255" fc="DC" />
  <DA name="yD" bType="VisString255" fc="DC" />
</DOType>
<DAType id="Vector_1">
  <BDA name="mag" type="AnalogueValue_1" bType="Struct" />
  <BDA name="ang" type="AnalogueValue_1" bType="Struct" />
</DAType>
<DAType id="Point_2">
  <BDA name="xVal" bType="FLOAT32" />
  <BDA name="yVal" bType="FLOAT32" />
</DAType>
<DAType id="Unit_3">
  <BDA name="SUnit" type="SUnitKind" bType="Enum" />
</DAType>
<DAType id="APCCancel_1">
  <BDA name="ctlVal" type="AnalogueValue_1" bType="Struct" />
  <BDA name="origin" type="Originator_1" bType="Struct" />
  <BDA name="ctlNum" bType="INT8U" />
  <BDA name="t" bType="Timestamp" />
  <BDA name="Test" bType="BOOLEAN" />
</DAType>
<DAType id="APCOperate_1">
  <BDA name="ctlVal" type="AnalogueValue_1" bType="Struct" />
  <BDA name="origin" type="Originator_1" bType="Struct" />
  <BDA name="ctlNum" bType="INT8U" />
  <BDA name="t" bType="Timestamp" />
  <BDA name="Test" bType="BOOLEAN" />
  <BDA name="Check" bType="Check" />
</DAType>
<DAType id="APCSelectWithValue_1">
  <BDA name="ctlVal" type="AnalogueValue_1" bType="Struct" />
  <BDA name="origin" type="Originator_1" bType="Struct" />
  <BDA name="ctlNum" bType="INT8U" />
  <BDA name="t" bType="Timestamp" />
  <BDA name="Test" bType="BOOLEAN" />
  <BDA name="Check" bType="Check" />
</DAType>
<DAType id="ENCSelectWithValue_1">

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<BDA name="ctlVal" type="Mod" bType="Enum" />
<BDA name="origin" type="Originator_1" bType="Struct" />
<BDA name="ctlNum" bType="INT8U" />
<BDA name="T" bType="Timestamp" />
<BDA name="Test" bType="BOOLEAN" />
<BDA name="Check" bType="Check" />
</DAType>
<DAType id="ENCOperate_1">
  <BDA name="ctlVal" type="Mod" bType="Enum" />
  <BDA name="origin" type="Originator_1" bType="Struct" />
  <BDA name="ctlNum" bType="INT8U" />
  <BDA name="T" bType="Timestamp" />
  <BDA name="Test" bType="BOOLEAN" />
  <BDA name="Check" bType="Check" />
</DAType>
<DAType id="ENCCancel_1">
  <BDA name="ctlVal" type="Mod" bType="Enum" />
  <BDA name="origin" type="Originator_1" bType="Struct" />
  <BDA name="ctlNum" bType="INT8U" />
  <BDA name="T" bType="Timestamp" />
  <BDA name="Test" bType="BOOLEAN" />
</DAType>
<DAType id="Originator_1">
  <BDA name="orCat" type="OriginatorCategoryKind" bType="Enum" />
  <BDA name="orIdent" bType="Octet64" />
</DAType>
<DAType id="AnalogueValue_1">
  <BDA name="f" bType="FLOAT32" desc="The value of f shall be the FLOAT representation of the measured value. f shall represent the technological value in SI units." />
</DAType>
<EnumType id="Mod">
  <EnumVal ord="1">on</EnumVal>
  <EnumVal ord="2">on-blocked</EnumVal>
  <EnumVal ord="3">test</EnumVal>
  <EnumVal ord="4">test/blocked</EnumVal>
  <EnumVal ord="5">off</EnumVal>
</EnumType>
<EnumType id="CtlModelKind">
  <EnumVal ord="0">status-only</EnumVal>
  <EnumVal ord="1">direct-with-normal-security</EnumVal>
  <EnumVal ord="2">sbw-with-normal-security</EnumVal>
  <EnumVal ord="3">direct-with-enhanced-security</EnumVal>
  <EnumVal ord="4">sbw-with-enhanced-security</EnumVal>
</EnumType>
<EnumType id="Beh">
  <EnumVal ord="1">on</EnumVal>
  <EnumVal ord="2">on-blocked</EnumVal>
  <EnumVal ord="3">test</EnumVal>
  <EnumVal ord="4">test/blocked</EnumVal>
  <EnumVal ord="5">off</EnumVal>
</EnumType>
<EnumType id="Health">
  <EnumVal ord="1">Ok</EnumVal>
  <EnumVal ord="2">Warning</EnumVal>
  <EnumVal ord="3">Alarm</EnumVal>
</EnumType>
<EnumType id="PhyHealth">
  <EnumVal ord="1">Ok</EnumVal>
  <EnumVal ord="2">Warning</EnumVal>
  <EnumVal ord="3">Alarm</EnumVal>
</EnumType>
<EnumType id="DERStateKind">
  <EnumVal ord="1">on but disconnected and not ready</EnumVal>
  <EnumVal ord="2">starting up</EnumVal>
  <EnumVal ord="3">disconnected and available</EnumVal>
  <EnumVal ord="4">disconnected and authorized</EnumVal>
  <EnumVal ord="5">synchronizing</EnumVal>
  <EnumVal ord="6">running</EnumVal>

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<EnumVal ord="7">stopping and disconnecting under emergency conditions</EnumVal>
<EnumVal ord="8">stopping</EnumVal>
<EnumVal ord="9">disconnected and blocked</EnumVal>
<EnumVal ord="10">disconnected and in maintenance</EnumVal>
<EnumVal ord="11">failed</EnumVal>
<EnumVal ord="98">Not applicable or not known</EnumVal>
</EnumType>
<EnumType id="PhaseKind">
<EnumVal ord="1">Single phase to neutral</EnumVal>
<EnumVal ord="2">Split phase</EnumVal>
<EnumVal ord="3">2-phase</EnumVal>
<EnumVal ord="4">3-phase delta</EnumVal>
<EnumVal ord="5">3-phase wye / 4-wires</EnumVal>
<EnumVal ord="6">3-phase wye / 5-wires</EnumVal>
<EnumVal ord="9">DC</EnumVal>
<EnumVal ord="98">Not applicable or not known</EnumVal>
</EnumType>
<EnumType id="DERUnitKind">
<EnumVal ord="1">Diesel / gas engine</EnumVal>
<EnumVal ord="2">Gas Turbine engine</EnumVal>
<EnumVal ord="3">PV</EnumVal>
<EnumVal ord="4">PV plus Storage</EnumVal>
<EnumVal ord="5">Lithium Ion Battery Storage</EnumVal>
<EnumVal ord="6">Fuel cell</EnumVal>
<EnumVal ord="7">Hydro generator</EnumVal>
<EnumVal ord="8">Wind turbine</EnumVal>
<EnumVal ord="9">Flow battery storage</EnumVal>
<EnumVal ord="10">Air compression storage</EnumVal>
<EnumVal ord="11">Flywheel storage</EnumVal>
<EnumVal ord="12">Capacitor storage</EnumVal>
<EnumVal ord="13">Vehicle-to-Grid (V2G)</EnumVal>
<EnumVal ord="50">Mixed, hybrid DER</EnumVal>
<EnumVal ord="98">Not applicable or not known</EnumVal>
</EnumType>
<EnumType id="OriginatorCategoryKind">
<EnumVal ord="0">not-supported</EnumVal>
<EnumVal ord="1">bay-control</EnumVal>
<EnumVal ord="2">station-control</EnumVal>
<EnumVal ord="3">remote-control</EnumVal>
<EnumVal ord="4">automatic-bay</EnumVal>
<EnumVal ord="5">automatic-station</EnumVal>
<EnumVal ord="6">automatic-remote</EnumVal>
<EnumVal ord="7">maintenance</EnumVal>
<EnumVal ord="8">process</EnumVal>
</EnumType>
<EnumType id="DERFunctionState">
<EnumVal ord="0">Not Available</EnumVal>
<EnumVal ord="1">Autonomous</EnumVal>
<EnumVal ord="2">Slave</EnumVal>
</EnumType>
<EnumType id="SIUnitKind">
<EnumVal ord="1" />
<EnumVal ord="10">rad</EnumVal>
<EnumVal ord="11">sr</EnumVal>
<EnumVal ord="2">m</EnumVal>
<EnumVal ord="21">Gy</EnumVal>
<EnumVal ord="22">Bq</EnumVal>
<EnumVal ord="23">°C</EnumVal>
<EnumVal ord="24">Sv</EnumVal>
<EnumVal ord="25">F</EnumVal>
<EnumVal ord="26">C</EnumVal>
<EnumVal ord="27">S</EnumVal>
<EnumVal ord="28">H</EnumVal>
<EnumVal ord="29">V</EnumVal>
<EnumVal ord="3">kg</EnumVal>
<EnumVal ord="30">ohm</EnumVal>
<EnumVal ord="31">J</EnumVal>

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<EnumVal ord="32">N</EnumVal>
<EnumVal ord="33">Hz</EnumVal>
<EnumVal ord="34">lx</EnumVal>
<EnumVal ord="35">Lm</EnumVal>
<EnumVal ord="36">Wb</EnumVal>
<EnumVal ord="37">T</EnumVal>
<EnumVal ord="38">W</EnumVal>
<EnumVal ord="39">Pa</EnumVal>
<EnumVal ord="40">s</EnumVal>
<EnumVal ord="41">m²</EnumVal>
<EnumVal ord="42">m³</EnumVal>
<EnumVal ord="43">m/s</EnumVal>
<EnumVal ord="44">m/s²</EnumVal>
<EnumVal ord="45">m³/s</EnumVal>
<EnumVal ord="46">m³/m²</EnumVal>
<EnumVal ord="47">M</EnumVal>
<EnumVal ord="48">kg/m</EnumVal>
<EnumVal ord="49">m³/s</EnumVal>
<EnumVal ord="50">A</EnumVal>
<EnumVal ord="51">V/m K</EnumVal>
<EnumVal ord="52">J/K</EnumVal>
<EnumVal ord="53">ppm</EnumVal>
<EnumVal ord="54">1s</EnumVal>
<EnumVal ord="55">rad/s</EnumVal>
<EnumVal ord="56">W/m²</EnumVal>
<EnumVal ord="57">J/m²</EnumVal>
<EnumVal ord="58">S/m</EnumVal>
<EnumVal ord="59">K/s</EnumVal>
<EnumVal ord="60">Pa/s</EnumVal>
<EnumVal ord="61">J/kg K</EnumVal>
<EnumVal ord="62">K</EnumVal>
<EnumVal ord="63">VA</EnumVal>
<EnumVal ord="64">Watts</EnumVal>
<EnumVal ord="65">VAr</EnumVal>
<EnumVal ord="66">phi</EnumVal>
<EnumVal ord="67">cos(phi)</EnumVal>
<EnumVal ord="68">Vs</EnumVal>
<EnumVal ord="69">V</EnumVal>
<EnumVal ord="70">As</EnumVal>
<EnumVal ord="71">A²</EnumVal>
<EnumVal ord="72">mol</EnumVal>
<EnumVal ord="73">A²</EnumVal>
<EnumVal ord="74">VAh</EnumVal>
<EnumVal ord="75">Wh</EnumVal>
<EnumVal ord="76">VArh</EnumVal>
<EnumVal ord="77">VHz</EnumVal>
<EnumVal ord="78">Hz/s</EnumVal>
<EnumVal ord="79">char</EnumVal>
<EnumVal ord="80">char/s</EnumVal>
<EnumVal ord="81">kgm²</EnumVal>
<EnumVal ord="82">dB</EnumVal>
<EnumVal ord="83">JWh</EnumVal>
<EnumVal ord="84">Ws</EnumVal>
<EnumVal ord="85">J/s</EnumVal>
<EnumVal ord="86">dBm</EnumVal>
<EnumVal ord="87">cd</EnumVal>
<EnumVal ord="88">deg</EnumVal>
</EnumType>
</DataTypeTemplates>
</SCL>
```

This Standard has been compiled by the **Comitato Elettrotecnico Italiano** and benefits from recognition under Law No. 186 of 1 March 1968.

Published by CEI, Comitato Elettrotecnico Italiano, Milan

Technical Committee

CT 57 - Power systems management and associated information exchange



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